37. Container Networking: Bridge, CNI, localhost Traps

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Phase: Phase 12 — Cloud, containers, and modern nets
Estimated time: 2 days
Slug: `37-container-net`

Learning objectives

- Explain docker bridge basics
- Publish ports correctly
- Debug DNS inside containers
- Avoid localhost confusion

Prerequisites

Complete prior lessons. Prefer a Linux VM for `ip`/`ss`/`nft` examples; macOS works for many `curl`/`dig` drills.

```
docker network ls
docker run --rm curlimages/curl:latest curl -4 -I https://example.com
docker run --rm -p 8080:80 nginx
curl -I http://127.0.0.1:8080

# inspect
docker network inspect bridge | head
nsenter tips / ip netns (advanced Linux)
```

`localhost` inside a container is the container, not the host. Use host gateway / service names / published ports.

Common mistakes

- –network host as default
- Assuming container DNS equals host DNS always
- Publishing all ports broadly

Exercises

Work in `exercises/37-container-net/`:

```
./test.sh
```

- **A:** make tests pass
- **B:** run 2–3 real commands from this lesson in your lab and note outputs
- **C:** rewrite answers from memory next day

Do not open `solutions/37-container-net/` until after an honest attempt.

Self-check

1. localhost in container means?
2. -p 8080:80 meaning?
3. docker network ls purpose?

Further reading

- https://docs.docker.com/network/